

Causation

Tim Houle, PhD

What is Causation?



What is Causation?

- Aristotle
 - Material cause
 - Formal cause
 - Efficient cause
 - Final cause
- Necessary Cause
 - The presence of *y* necessarily implies the *prior* occurrence of *x*
- Sufficient Cause
 - The presence of *x* necessarily implies the *subsequent* occurrence of *y*

Level of evidence (LOE)	Description
Level I	Evidence from a systematic review or meta-analysis of all relevant RCTs (randomized controlled trial) or evidence-based clinical practice guidelines based on systematic reviews of RCTs or three or more RCTs of good quality that have similar results.
Level II	Evidence obtained from at least one well-designed RCT (e.g. large multi-site RCT).
Level III	Evidence obtained from well-designed controlled trials without randomization (i.e. quasi-experimental).
Level IV	Evidence from well-designed case-control or cohort studies.
Level V	Evidence from systematic reviews of descriptive and qualitative studies (meta-synthesis).
Level VI	Evidence from a single descriptive or qualitative study.
Level VII	Evidence from the opinion of authorities and/or reports of expert committees.



Causation

Methods for Gathering Evidence

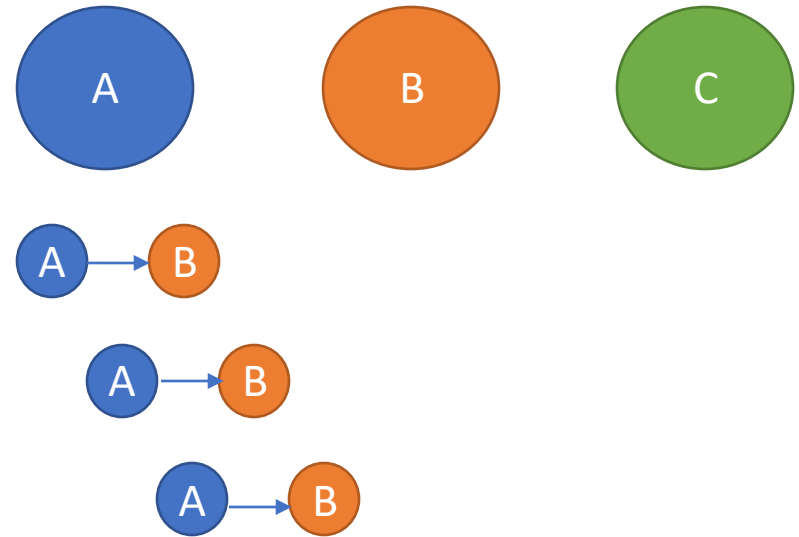
Research Design

- Systematic review of randomized controlled trials
- Experiment (e.g., RCT)
- Quasi-experiment (e.g., interrupted time-series)
- Observational (e.g., cohort, cross-sectional)
- Pre-experimental (e.g., pretest-posttest)
- Natural experiment
- Case-study (series)

Which allows for 'causal inference'?

David Hume (1711 – 1776)

- Causation is elusive
- We can only observe a series of events
- Constant Conjunctions
 - Frequentism
- Inductive Skepticism



Correlation *does not* imply
causation...

But

Causation *does (often)* imply
correlation...

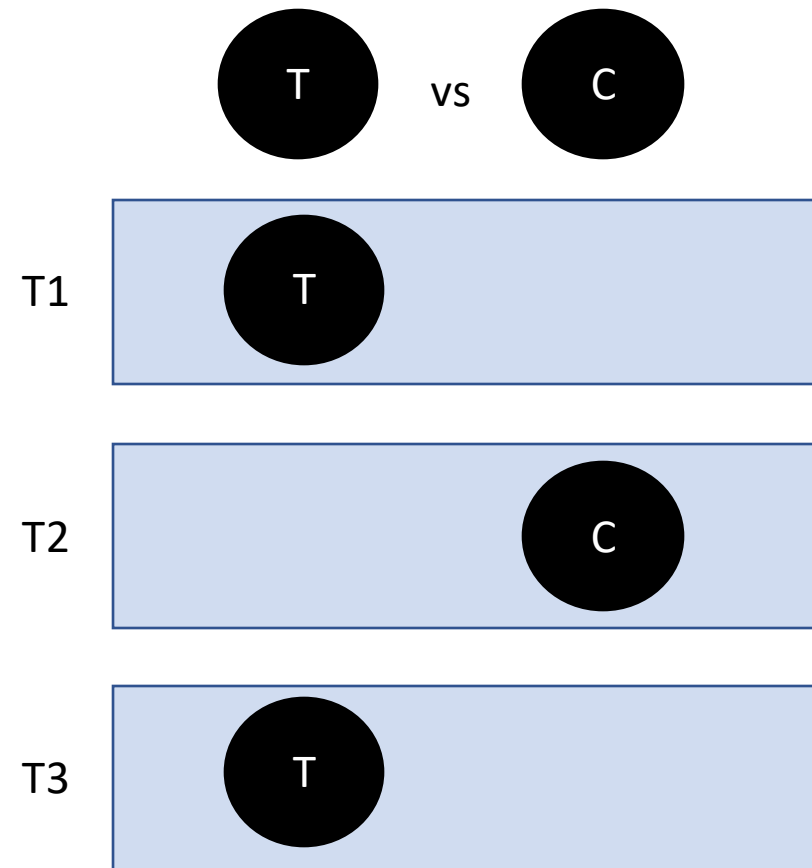
Bradford Hill Criteria (1965)

- Strength: Effect size
- Consistency: Reproducibility
- Specificity: Specific effect
- Temporality: Cause precedes effect
- Biological Gradient: Dose-response relationship
- Plausibility: theoretically sound mechanism
- Coherence: Between sources of evidence
- Experiment: Available experimental evidence
- Analogy: Similarities with other phenomena

Rubin Causal Model

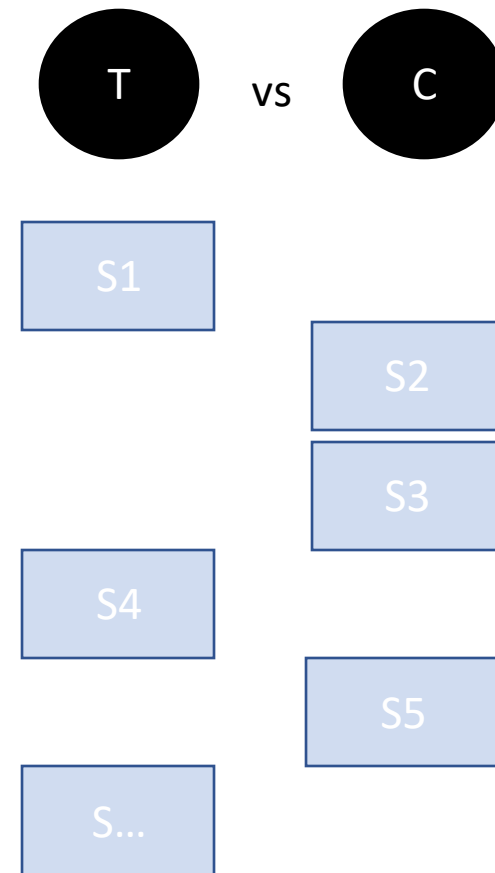
Neyman-Rubin Causal Model (1923)

- Potential Outcomes
 - $Y_t(Tx) - Y_t(Con)$
- Fundamental Problem of Causation
- Unverifiable Assumptions
 - Exchangeability
 - Constancy
 - Permanence



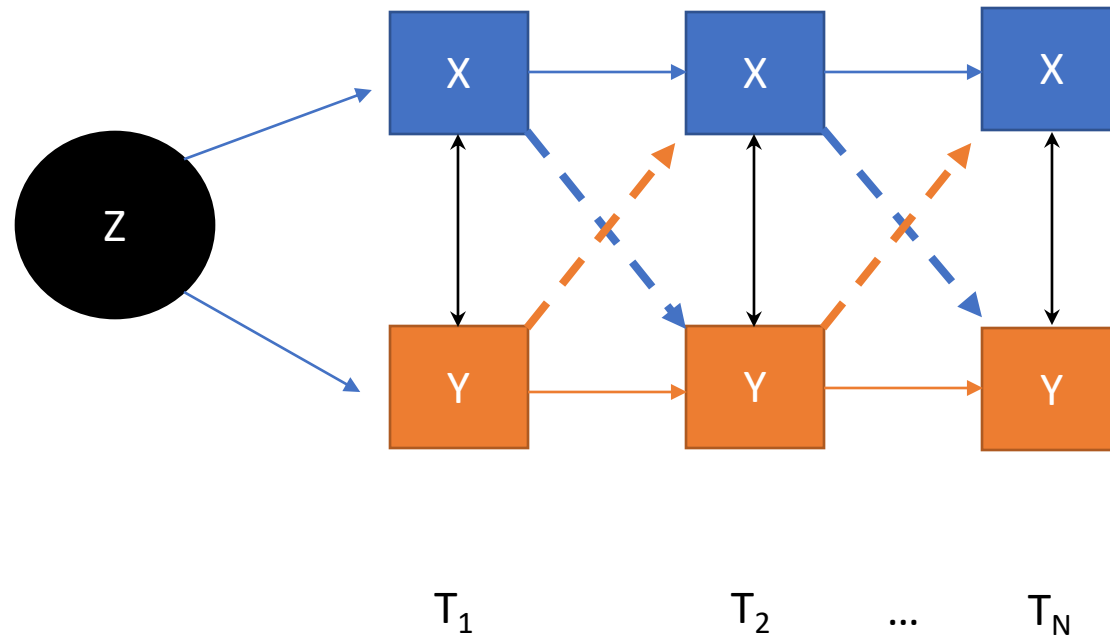
Rubin Causal Model

- Potential Outcomes
 - $Y_t(\text{Tx}) - Y_t(\text{Con})$
- Randomized Controlled Trial
 - Randomly assign participants to intervention
 - Expected value of characteristics
 - $E(\text{confounder}_T) = E(\text{confounder}_C)$



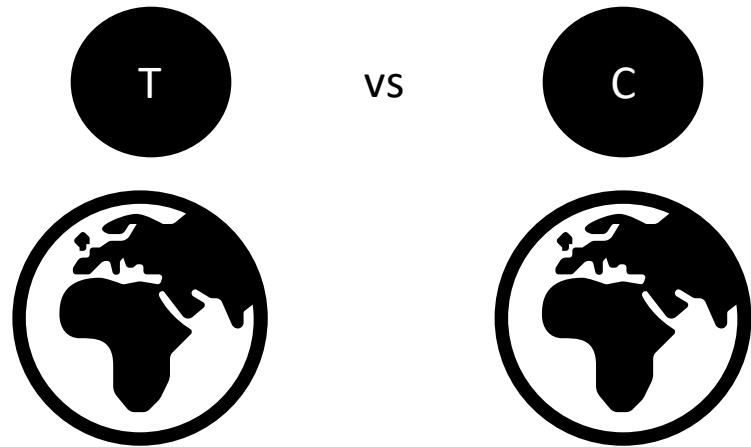
Granger Causality

- Cross-lagged panel model (Campbell [1963]; Kenny [1970])
- Predictive causality



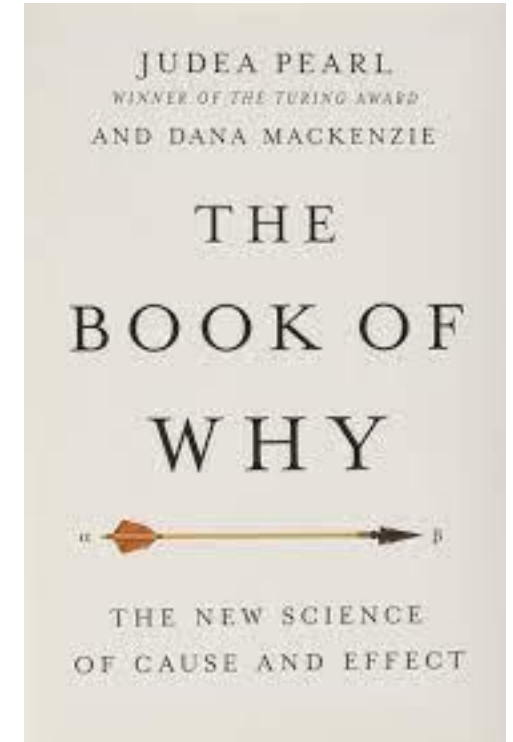
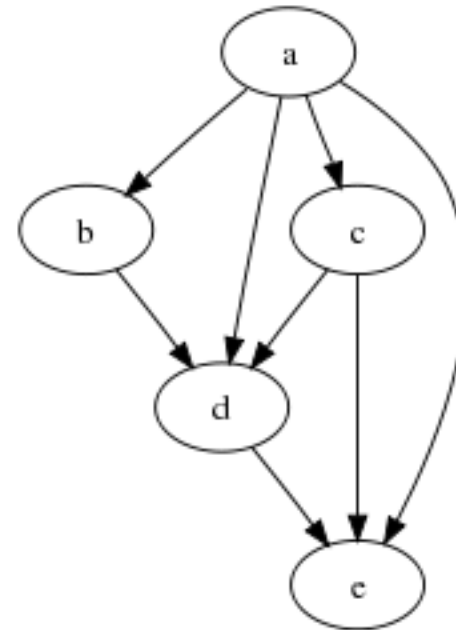
Lewis (1973) Theory of Counterfactuals

- “If A had not occurred, C would not have occurred”
- Possible world semantics

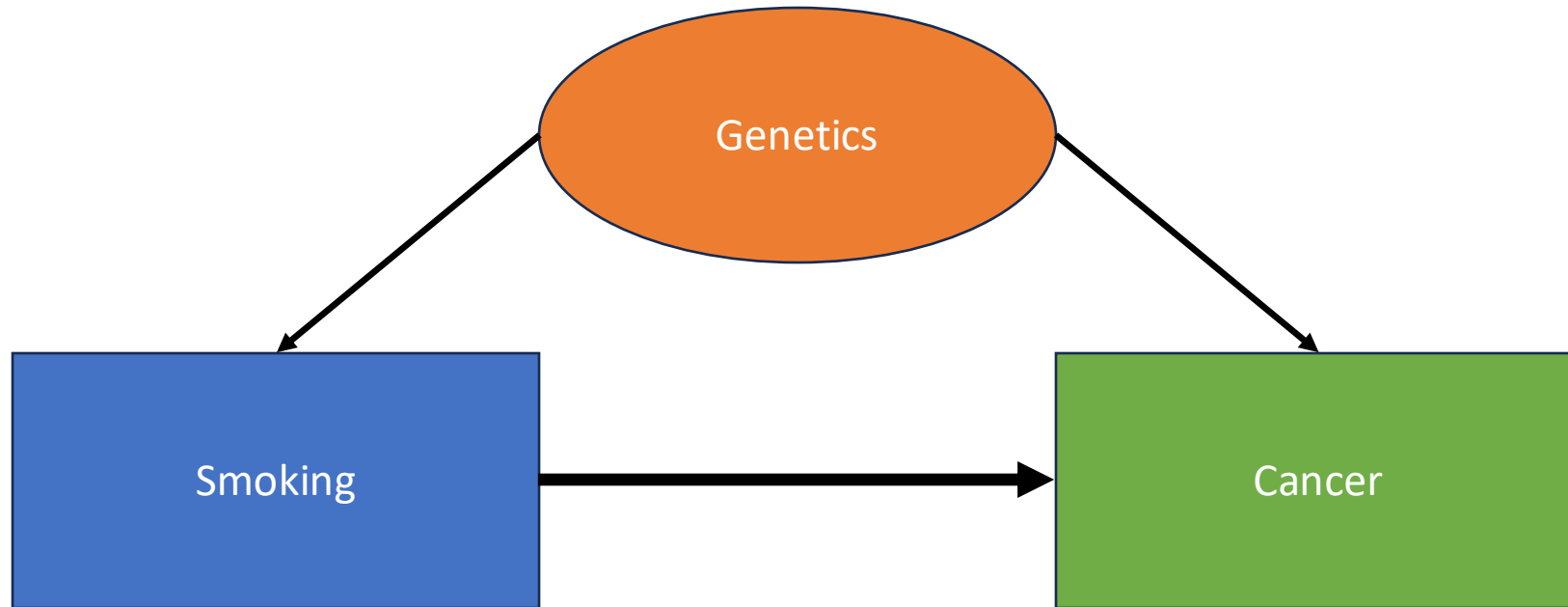


Judea Pearl's Causal Revolution

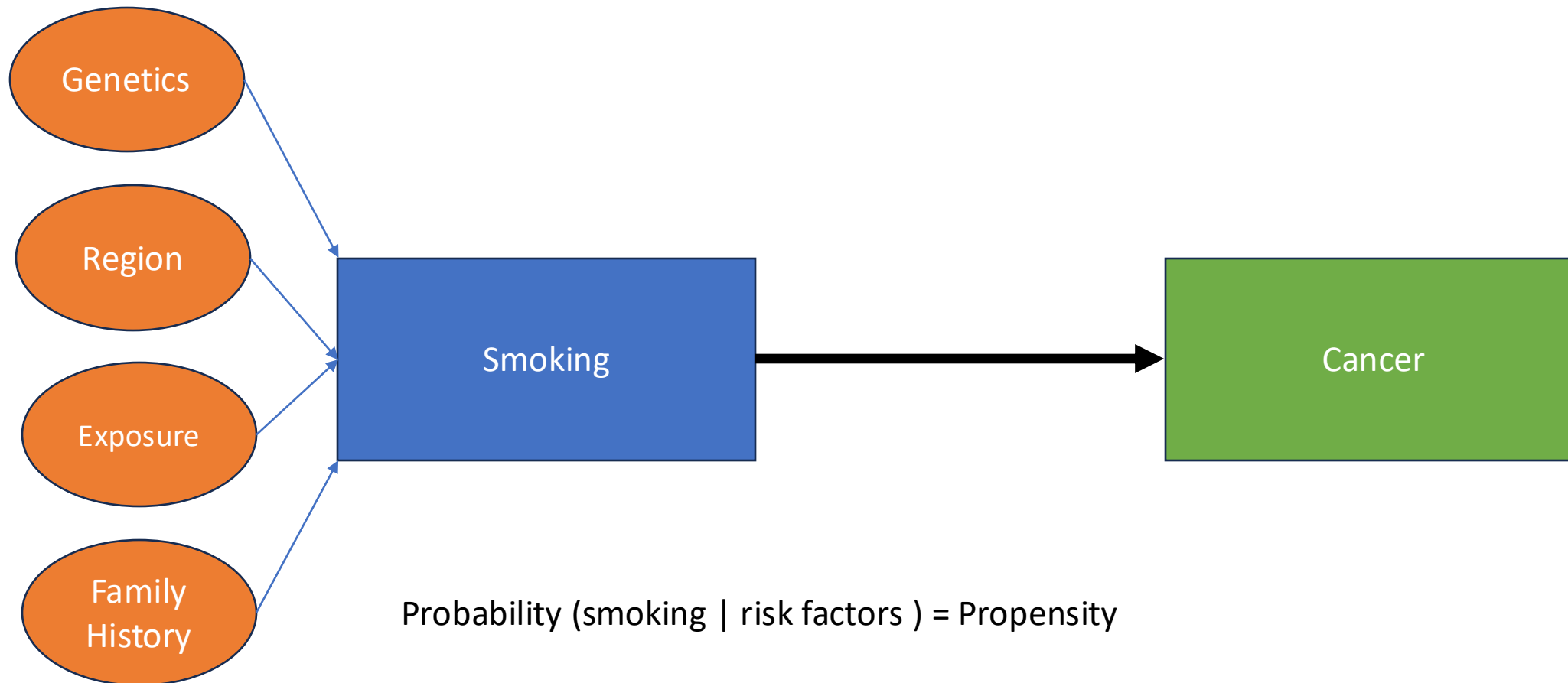
- Structural Causal Model
 - Directed Acyclic Graph (DAG)
- The 'Do' Operator
 - Synthetic 'intervention'
- Structural Equation Modeling (SEM)



Case Study: Smoking and Cancer



Case Study: Smoking and Cancer



Propensity Scores

<u>Non-Smokers</u>	<u>Smokers</u>
0.043	0.022
0.156	0.159
0.267	0.259
0.568	0.567
...	...
0.943	0.943

$$P(\text{Cancer}_{\text{non-smoker}}) = P(\text{Cancer}_{\text{smoker}})$$

2% vs. 8%

Propensity Methods

- Matching
- Stratification
- Adjustment / Weighting

Case Study 2:

The Risk Associated with Aprotinin in Cardiac Surgery

Dennis T. Mangano, Ph.D., M.D., Iulia C. Tudor, Ph.D., and Cynthia Dietzel, M.D. for the Multicenter Study of Perioperative Ischemia Research Group and the Ischemia Research and Education Foundation*

BACKGROUND

The majority of patients undergoing surgical treatment for ST-elevation myocardial infarction receive antifibrinolytic therapy to limit blood loss. This approach appears counterintuitive to the accepted medical treatment of the same condition — namely, fibrinolysis to limit thrombosis. Despite this concern, no independent, large-scale safety assessment has been undertaken.

Table 2. Results of Multivariable Logistic Regression for the Renal Composite Outcome in 4374 Patients.*

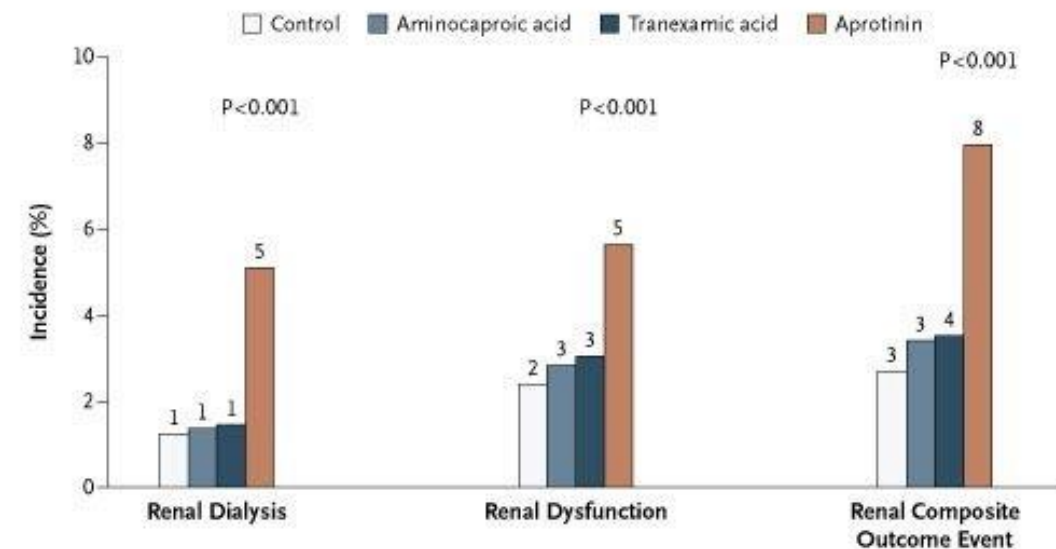
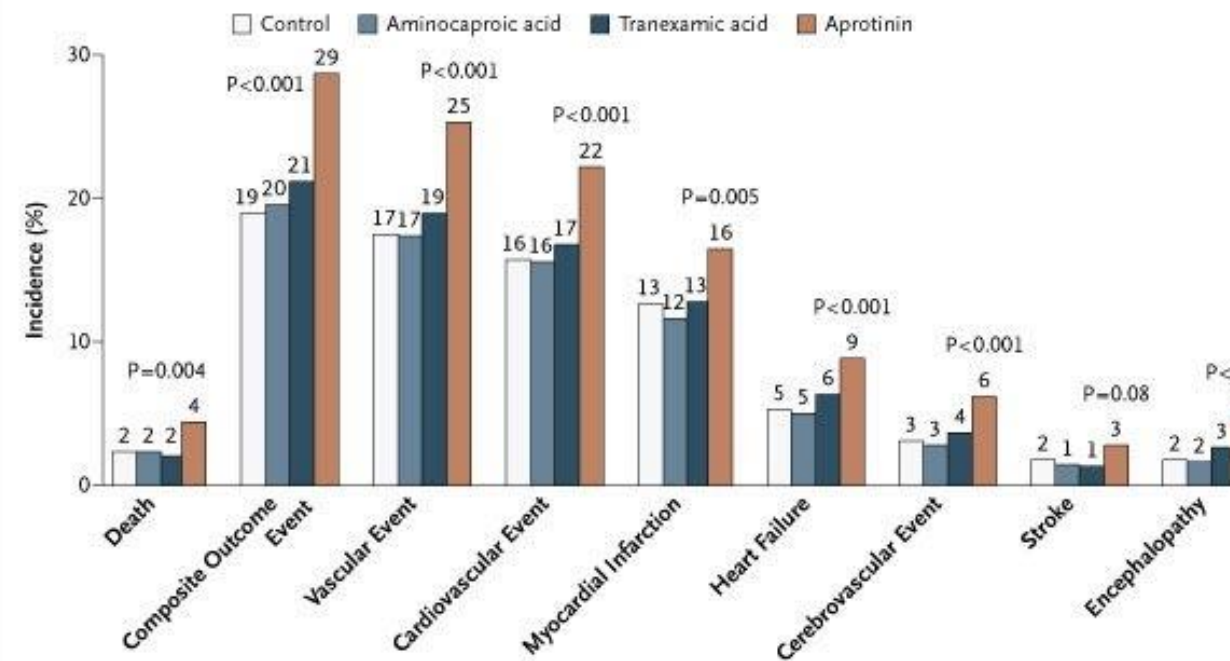
Risk Factor	Analysis in Presence of Covariates without Propensity Adjustment†		Analysis in Presence of Covariates with Propensity Adjustment‡	
	Odds Ratio (95% CI)	P Value	Odds Ratio (95% CI)	P Value
Aprotinin vs. control	2.52 (1.66–3.82)	<0.01	2.41 (1.49–3.90)	<0.001
Aminocaproic acid vs. control	1.03 (0.61–1.76)	0.91	0.84 (0.44–1.58)	0.58
Tranexamic acid vs. control	1.25 (0.74–2.13)	0.40	1.23 (0.68–2.21)	0.49
Propensity score	—	—	1.03 (0.97–1.11)	0.33
Complex vs. primary surgery	1.55 (1.12–2.16)	0.009	1.47 (1.02–2.13)	0.04
History of renal disease	2.50 (1.76–3.57)	<0.001	2.53 (1.70–3.75)	<0.001
Creatinine >1.3 mg/dl on admission§	2.71 (1.91–3.87)	<0.001	3.12 (2.11–4.60)	<0.001
Heart failure on admission	2.33 (1.68–3.24)	<0.001	2.64 (1.84–3.80)	<0.001
History of angina	0.57 (0.36–0.89)	0.01	0.58 (0.35–0.96)	0.03
History of liver disease	0.35 (0.18–0.68)	0.002	0.28 (0.13–0.61)	0.001
History of intravenous drug use	3.29 (1.08–10.04)	0.04	2.98 (0.82–10.84)	0.10
Preoperative nitrate administration	1.95 (1.29–2.93)	0.001	2.15 (1.36–3.40)	0.001
Preoperative inotrope administration	2.31 (1.45–3.67)	<0.001	2.36 (1.42–3.92)	<0.001
Preoperative administration of ACE inhibitor	1.38 (1.00–1.91)	0.05	1.57 (1.10–2.24)	0.01
Intraoperative transfusion of fresh frozen plasma	2.51 (1.72–3.65)	<0.001	2.40 (1.58–3.66)	<0.001
Intraoperative transfusion of red cells	1.64 (1.15–2.34)	0.007	1.71 (1.16–2.52)	0.007

* CI denotes confidence interval, and ACE angiotensin-converting enzyme.

† Excluded were 43 patients with missing values for at least one of the risk factors in the model, including the covariates. The Hosmer–Lemeshow goodness-of-fit chi-square test statistic was 10.3 ($P=0.24$). The C-index for the model was 0.84. Direct comparisons of treatment groups by contrast functions on model parameters demonstrated increased risk with aprotinin as compared with aminocaproic acid (odds ratio, 2.44; 95 percent confidence interval, 1.54 to 3.86; $P<0.001$) and as compared with tranexamic acid (odds ratio, 2.01; 95 percent confidence interval, 1.27 to 3.18; $P=0.003$). There was no difference in risk associated with the use of aminocaproic acid as compared with tranexamic acid.

‡ Excluded were 410 patients with missing values for at least one of the covariates or the propensity score. The Hosmer–Lemeshow goodness-of-fit chi-square test statistic was 7.17 ($P=0.52$). The C-index for the model was 0.86. Direct comparisons of treatment groups by contrast functions on model parameters demonstrated increased risk with aprotinin as compared with aminocaproic acid (odds ratio, 2.88; 95 percent confidence interval, 1.69 to 4.90; $P<0.001$) and as compared with tranexamic acid (odds ratio, 1.96; 95 percent confidence interval, 1.19 to 3.23; $P=0.009$). There was no difference in risk associated with the use of aminocaproic acid as compared with tranexamic acid.

§ To convert the value for creatinine to micromoles per liter, multiply by 88.4.

A**B**

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More Reading

- Simpson's paradox
 - Associations differ depending on the variables in the model, and their relationships
 - <https://www.britannica.com/topic/Simpsons-paradox/Problem-of-Causality>
- Rubin's Causal Model
 - https://link.springer.com/chapter/10.1057/9780230280816_28
- Judea Pearl's Twitter Feed
 - https://twitter.com/yudapearl?ref_src=twsrc%5Egoogle%7Ctwcamp%5Eserp%7Ctwgr%5Eauthor